SHIKANG PENG

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EDUCATION

2024- Ph.D., Psychology – Computational Cognitive Neuroscience
University of Toronto
Advisors: Drs. Brian Levine & Bradley Buchsbaum

2023-24 M.A., Social Sciences – Psychology
University of Chicago
Advisor: Dr. Wilma Bainbridge
Thesis: Predictability of memorability neural pattern and attention network in subsequent memory retrieval

2019-23 B.Sc., Psychology with Honors | First Class Honors
University of Alberta
Advisor: Dr. Peter Dixon
Thesis: Text-based and memory-based metrics of cognitive coupling

PREPRINTS

Peng, S., Levine, B., & Buchsbaum, B. R. (Under review). Naturalistic recall preserves temporal specificity despite cortical feature reorganization. *bioRxiv.* https://doi.org/10.64898/2026.11.03.763513

Peng, S., Buchsbaum, B. R., & Levine, B. (Under review). Hippocampal long-axis organization reflects individual differences in autobiographical remembering.

St Jacques, P., Feminella, J., & **Peng, S.** (Under review). Seeing events in 3D enhances scene memory and vividness. *PsyArXiv*. https://doi.org/10.31234/osf.io/z9thr

PUBLICATIONS

2. **Peng, S.**, & Bainbridge, W. A. (2026). Image memorability predicts social media virality and externally-associated commenting. *Computers in Human Behavior, 174,* 108799. https://doi.org/10.1016/j.chb.2025.108799
1. **Peng, S.**, & Dixon, P. (2024). Text-based and memory-based metrics of cognitive coupling. *Canadian Journal of Experimental Psychology*, *78*(4), 222-229. https://doi.org/10.1037/cep0000349

CONFERENCE PRESENTATIONS (* denotes presenting author)

Oral Presentations

2. **Peng, S.**, Buchsbaum, B. R.*, & Levine, B. (2026, Oct). On the vividability of things and scenes. *37th Annual Meetings of Memory Disorder Research Society (MDRS)*, Toronto, ON, CA.
1. **Peng, S.***, Levine, B, & Buchsbaum, B. R. (2026, May). Beyond reinstatement: Transformation of cortical feature representations during memory recall. *Toronto Area Memory Group Conference (TAMeG)*, Toronto, ON, CA.

Poster Presentations

7. **Peng, S.***, Utochkin, I. S., Levine, B., & Bainbridge, W. A. (2026, Nov). Behavioral change is not a unique signature of change in memory organization. *Annual Meetings of Psychonomic Society*, San Diego, CA, US.
6. **Peng, S.***, Buchsbaum, B. R., & Levine, B. (2026, Mar). Hippocampal-cortical connectivity captures traits in autobiographical memory. *Annual Meetings of Cognitive Neuroscience Society (CNS)*, Vancouver, BC, CA.
5. **Peng, S.***, & Bainbridge, W. A. (2025, May). Image memorability predicts widespread virality on social media. *Annual Meetings of Vision Science Society (VSS)*, St. Pete Beach, FL, US.
4. **Peng, S.***, & Levine, B. (2025). Cortical thickness and volume differences in individuals with severely-deficient autobiographical memory. *Annual Meetings of Cognitive Neuroscience Society (CNS)*, Boston, MA, US.
3. Feminella, J.*, **Peng, S.**, & St Jacques, P. (2024, Apr). Dimensionality affects memory for events in immersive virtual reality. *Annual Meetings of Cognitive Neuroscience Society (CNS)*, Toronto, ON, CA.
2. **Peng, S.***, & Dixon, P. (2022, July). Mind wandering and temporal focus on task switching. *Annual Meetings of Canadian Society for Brain, Behavior and Cognitive Science (CSBBCS)*, Halifax, NS, CA.
1. **Peng, S.***, & Dixon, P. (2022, Apr). Involvement of mind wandering in the inconsistent contradiction effect during narrative reading. *Royce Harder Conference*, Edmonton, AB, CA.

GRANTS, AWARDS, & HONORS

2026 Travel Award, Department of Psychology, University of Toronto ($500)
2026 Data Access Grant, Data Sciences Institute ($10,000)
2026 Men's Service Group Graduate Fellowship, Baycrest ($2,250)
2025-26 Jack & Rita Catherall Travel Fund, Baycrest ($500)
2025 Max & Ruth Wiseman Graduate Fellowship, Baycrest ($2,300)
2023 Maroon Research Scholarship, University of Chicago ($30,000)
2023 M.A. Director Scholarship, University of Chicago ($2,000)
2023 Dean's Silver Medal in Science, University of Alberta
2022 Clare Patershuk Travel Award, University of Alberta ($750)
2021-23 Dean's Honor Roll in Science, University of Alberta

TEACHING

Course instructor

2026 Human Memory – PSY372, Department of Psychology, University of Toronto

Teaching assistant

2026 Human Memory – PSY372, Department of Psychology, University of Toronto
2025 Statistics II – PSY202, Department of Psychology, University of Toronto
2025, 26 Introductory Psychology – PSY100, Department of Psychology, University of Toronto
2024,25,27 Statistics I – PSY201, Department of Psychology, University of Toronto

MENTORING

2026 Leo Zhang (Psychology undergraduate, summer program, University of Waterloo)
2025 Rivka Zeng-van Klei (Neuroscience undergraduate, summer program, York University)

ACADEMIC COMMUNITY SERVICES

Ad hoc review

Acta Psychologica, Journal of Experimental Psychology: General

Other services

2026 *Graduate Advisor*, Inkblot: Undergraduate psychology journal, University of Toronto
2026 *Research Area Representative (Cognitive Neuroscience)*, UofT graduate recruitment symposium
2025 *Planning Committee*, Rotman research institute annual research conference
2024- *Member*, Cognitive science research community (CoRC), University of Toronto
2024-26 *Fundraising & Social Committee*, Annual meetings of Toronto area memory group (TAMeG)

RESEARCH SOFTWARE

Peng, S. (2025). LongiGate: An automatic gatekeeper for online longitudinal studies. *Zenodo.* https://doi.org/10.5281/zenodo.15253316

TECHNICAL SKILLS

Coding Python (PsychoPy, PyTorch), R, MATLAB, BASH, JavaScript (jsPsych), HTML/CSS
Compute HPC/SLURM, Cloudflare (ETL workflow)
Software FreeSurfer, SPM12, AFNI, FSL, Pavlovia, Prolific, Qualtrics, Vuze VR Studio, Unity
Imaging Functional/structural MRI, EEG/ERP

CERTIFICATES

2023 Computational Neuroscience, Neuromatch Academy

REFERENCES

Brian Levine, Ph.D., C. Psych., ABPP-cn, FRSC
Professor, University of Toronto | Senior Scientist, Rotman Research Institute
Ph.D. Thesis Supervisor

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Bradley Buchsbaum, Ph.D.
Professor, University of Toronto | Senior Scientist, Rotman Research Institute
Ph.D. Thesis Co-supervisor
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Wilma Bainbridge, Ph.D.
Associate Professor, University of Chicago
Master's Thesis Main Supervisor
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